
Riverbend Shipping

User Manual

A WooCommerce shipping calculator for Riverbend Aquatics:
livestock delivery, dry-goods delivery, very-large-item charges,
cooler boxes, heat packs and an itemised breakdown on every order.

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Contents

1	Introduction	4
1.1	What this plugin does	4
1.2	What it does not do	4
1.3	How it is put together	4
2	Installation and first-time setup	6
2.1	Requirements	6
2.2	Installing the plugin	6
2.3	Adding it to a shipping zone	6
2.4	First-run checklist	7
3	Settings reference	8
3.1	Shipping rates	8
3.2	Free shipping	8
3.3	Cooler boxes and heat packs	9
3.3.1	The two season modes	10
3.4	Delivery areas	10
3.5	Product shipping types	10
3.6	Checkout options and notices	11
4	Setting up products	13
4.1	The three things a product needs	13
4.2	The fields the plugin adds	13
4.2.1	What happens when they are not set	13
4.3	Bulk editing	14
4.4	A practical order of work	14
5	How the price is calculated	15
5.1	Step 0: which area is this going to?	15
5.2	Step 1: sorting the cart	16
5.3	Step 1a: does this order earn free shipping?	16
5.4	Step 2: dry-goods delivery	17
5.5	Step 3: very-large items	18

5.6	Step 4: livestock delivery and packaging	18
5.6.1	Delivery	18
5.6.2	Bags	18
5.6.3	Cooler boxes	19
5.6.4	Charges	19
5.7	Step 5: gathering the breakdown	20
5.8	Step 6: the whole-order rules	20
5.9	Where the money ends up at checkout	21
5.10	Worked example: a mixed order	22
5.11	Worked example: a small local order	22
6	What customers and staff see	24
6.1	On the cart and checkout	24
6.1.1	The delivery options	24
6.1.2	The breakdown under the rate	24
6.1.3	The packaging fees	24
6.1.4	The notices	24
6.2	On the order	24
6.2.1	The customer's view	25
6.2.2	The shop's view	25
6.2.3	Why the numbers never change retrospectively	25
7	Routine maintenance	26
7.1	Changing a rate	26
7.2	Turning heat-pack season on and off	26
7.3	Adjusting the local delivery area	26
7.4	Adding a new product	26
7.5	Reading the log	26
7.6	Checking a quote by hand	27
7.7	Upgrading the plugin	27
8	Troubleshooting	28
9	Known limitations and open decisions	30
9.1	Decisions made on your behalf	30
9.2	Still outstanding	30
9.3	Collection appears twice if Local Pickup is also enabled	31

10 Technical reference	32
10.1 What is in the distributed package	32
10.2 Building a release	32
10.3 Development tooling	32
10.4 Design rules worth preserving	32
10.5 The manual-adjustment filter	33
10.6 Verification status	33
A Settings quick reference	34
B Data the plugin stores	36
B.1 Product meta	36
B.2 Order meta	36
C Glossary	37

Introduction

1.1 What this plugin does

Riverbend Shipping replaces WooCommerce's flat-rate shipping with a calculator built around how an aquatics shop actually dispatches an order. It knows the difference between a bag of shrimp, a glass aquarium and a tube of glue, and prices each of them the way Riverbend Aquatics prices them today.

For any given cart it works out:

- **Dry-goods delivery:** one flat charge for the ordinary items in the order.
- **The very-large-item charge:** charged per very-large item, *instead of* the dry-goods rate, not in addition to it.
- **Livestock delivery:** at a local or a national rate, depending on the delivery postcode.
- **Cooler boxes:** how many insulated boxes the livestock needs, from the number and size of packing bags, and what they cost.
- **Heat packs:** one per cooler box, but only while heat-pack season is on.
- **The free-shipping discount:** whether the order has earned free dry-goods delivery, and what that is worth.
- **Collection:** an order collected from the Riverbend store pays no transport, but still pays for its packaging.

Every one of those components is stored on the order and shown to the customer, so the shipping figure on an invoice can always be explained.

1.2 What it does not do

- It does not calculate VAT. Every rate you enter is *excluding* VAT, and WooCommerce adds tax according to your own tax settings.
- It does not block checkout on non-dispatch days. It shows the dispatch information and lets the customer decide.
- It does not price an outlying delivery. It charges the national rate, flags the order, and leaves the surcharge to a conversation with the customer.
- It does not reclassify your catalogue. A product with no shipping class keeps being charged the ordinary dry-goods rate, exactly as it is today.

1.3 How it is put together

The plugin is two layers, and the split is worth knowing about because it explains where to look when something needs changing.

Layer	What lives there
The engine includes/	Every pricing rule. Contains no WordPress code at all: no options, no database, no clock. It takes a cart plus a set of settings and returns a priced breakdown. This is what makes the pricing testable without a shop.
The adapter wordpress/	The only code that knows what WooCommerce is. It gathers the cart, hands it to the engine, and puts the answer back where WooCommerce expects it: as a shipping rate, as cart fees, and as order meta.

Note

The practical consequence: changing a *price* is a settings change in wp-admin. Changing a *rule* is a change to the engine. Nothing about the pricing is hidden in a theme or a template.

Installation and first-time setup

2.1 Requirements

Requirement	Minimum
WordPress	6.4
WooCommerce	8.0
PHP	8.2
Order storage	Works with both legacy post storage and High-Performance Order Storage (HPOS).

The plugin has **no runtime dependencies**. There is nothing to build, no Composer install to run on the server, and no `vendor` folder in the distributed package.

2.2 Installing the plugin

1. In wp-admin, go to **Plugins** → **Add New** → **Upload Plugin**.
2. Choose `riverbend-shipping-0.1.0.zip` and select **Install Now**.
3. Select **Activate Plugin**.

Activation seeds every setting with its default value, so the settings screen shows real numbers immediately rather than a page of empty fields. Seeding never overwrites a value that already exists, so deactivating and reactivating the plugin will not undo your configuration.

Test on a staging site first

The plugin changes what customers are charged and what delivery options they are offered. Install it on a staging copy of the live site, work through Chapter 5 against real products, and only then move it to production.

2.3 Adding it to a shipping zone

Installing the plugin does not by itself put it in front of a customer. WooCommerce only offers a shipping method that has been added to a zone.

1. Go to **WooCommerce** → **Settings** → **Shipping** → **Zones**.
2. Open the zone that covers the country (or create one).
3. Select **Add shipping method** and choose **Riverbend Shipping**.
4. Save.

Repeat for every zone that should use these rates. If a zone has other methods on it (Flat rate, Free shipping, Local pickup), the customer will see those alongside Riverbend Shipping. See Section 9.3

about Local Pickup in particular.

2.4 First-run checklist

Work through this list before the plugin goes anywhere near a real customer.

Check	Where
☐ Every rate is entered <i>excluding</i> VAT.	Settings, Shipping rates
☐ The free-shipping threshold and its exclusion list are right.	Settings, Free shipping
☐ Cooler-box capacity and charges match what the shop actually does.	Settings, Cooler boxes
☐ Heat-pack season is set to the correct mode and state.	Settings, Cooler boxes
☐ The local-delivery postcode list has been corrected.	Settings, Delivery areas
☐ Every shipping-class slug in the translation list actually exists.	Products → Shipping classes
☐ Livestock products have a units-per-bag and a bag size set.	Product edit / bulk edit
☐ Free-shipping exclusions are ticked on the products that need them.	Product edit / bulk edit
☐ The four customer notices read the way you want them to.	Settings, Checkout notices
☐ A test order has been placed and its breakdown checked.	Front end

The VAT trap

Every rate on this screen is *excluding* VAT. A shop whose published prices include it has to convert before typing. Enter 160 where R160 was meant to be the inclusive figure, with shipping set to taxable, and every customer immediately starts paying R184.00 instead. Decide which of the two figures you are entering before you type it, not afterwards.

Settings reference

Every setting lives in one place:

WooCommerce → Settings → Shipping → Riverbend Shipping

There is no separate admin menu for this plugin. The screen is built with WooCommerce's own settings system, so it looks and saves like the rest of the store.

Settings are stored one row per value, each key prefixed `riverbend_shipping_`. A single bad value therefore cannot make the rest unreadable. If the saved settings cannot be validated at all, the plugin falls back to the shipped defaults and writes a warning to the log rather than breaking a checkout.

3.1 Shipping rates

All four figures exclude VAT.

Field	Default	Meaning
Local livestock delivery	R120.00	Livestock delivery to a postcode on the local list.
Livestock delivery elsewhere in the country	R280.00	Livestock delivery anywhere else, including outlying addresses.
Normal dry-goods delivery	R160.00	One flat charge for the whole dry-goods part of the order, however many items it contains.
Very large product delivery	R320.00	Charged <i>per very-large item</i> , instead of the dry-goods rate, not as well as it.

Note

There is currently one dry-goods rate and one very-large rate for the whole country. Only livestock has a local/national split. If a cheaper local tier is ever wanted for the other two, it is a settings change plus one line of code, with no change to the pricing logic itself.

3.2 Free shipping

Field	Default	Meaning
Free dry-goods shipping from	R2 500.00	Measured on the merchandise value <i>before</i> coupon discounts, excluding shipping and packaging.

Field	Default	Meaning
An order exactly on the threshold qualifies	ticked	Ticked: R2 500.00 exactly qualifies. Unticked: the order must exceed it.
Shipping types that block free shipping	all five non-dry-goods types	Any of these anywhere in the cart disqualifies the whole order, however much it is worth.
Also exclude products heavier than (kg)	4	A product over this weight blocks free shipping for the whole order. Set to 0 to switch the rule off.

The five types selected by default are: *Very large dry goods*, *Livestock and plants*, *Collection only*, *Shipping quote required* and *Livestock shipping unavailable*. In other words, only an order made entirely of ordinary dry goods can earn free shipping.

The weight rule fires unevenly

The weight exclusion can only act on products that actually carry a weight. On a catalogue where weights are patchy it therefore applies to some heavy items and not others, which looks arbitrary from the outside. It is a setting rather than a fixed rule precisely so that keeping it, or switching it off with 0, is a deliberate decision rather than an accident.

3.3 Cooler boxes and heat packs

Packaging applies to livestock orders whether or not a courier is involved, **including collection from the store**.

Field	Default	Meaning
Standard bags per cooler box	4	How many standard packing bags fit in one cooler box.
Space used by a large bag	2	In standard-bag places. A large bag currently takes two.
Charge per cooler box	R60.00	What the customer pays per box.
Charge per heat pack	R25.00	What the customer pays per heat pack.
Heat-pack season	by hand	Either Switch on and off by hand or Between two dates each year.
Heat-pack season is on now	unticked	Only consulted when the mode is by hand.
Season starts (MM-DD)	05-01	Only consulted in date mode.
Season ends (MM-DD)	08-31	Only consulted in date mode.
Packaging stays payable when transport is free	ticked	Untick only if a free-transport order should also stop paying for its cooler boxes.

3.3.1 The two season modes

By hand is the default and nothing turns itself on. You tick *Heat-pack season is on now* when winter starts and untick it when it ends. The start and end date fields are ignored.

Between two dates compares month and day only, so a window set once keeps working every year. A window that runs backwards, for example 11-01 to 02-28, correctly wraps across the new year rather than matching nothing.

The date is the shop's own date, not the server's, so a local winter setting does not switch over at UTC midnight.

3.4 Delivery areas

Postcodes may be written one per line or separated by commas or semicolons. Ranges work: 2400–2599. Spaces around the dash are fine. Anything not listed is treated as elsewhere in the country.

Field	Default	Meaning
Local delivery postcodes	2100–2160, 2400–2599, 2700–2999, 3000–3099	Gets the local livestock rate.
Outlying postcodes	empty	Priced at the national rate, but the order is flagged so an additional fee can be confirmed with the customer before dispatch.

The shipped postcodes are a starting point, not an answer

The default list is a rough boundary around the Riverbend metro: the coastal strip, the northern suburbs, the southern line and the city centre. It ships as a placeholder so the setting is never empty, and it **needs checking against the areas the shop actually drives to** before the plugin goes live. A postcode in the list is a promise to deliver there at the local rate.

A postcode cannot be on both lists. The settings screen refuses to save an overlap and tells you which entries clash.

3.5 Product shipping types

This is where your WooCommerce shipping classes are translated into the meanings the calculator understands. One rule per line, written as `class-slug = meaning`. Blank lines and lines starting with # are ignored.

Meaning	What it does
dry_goods	Ordinary item. Goes in the dry-goods bucket at the flat rate.
very_large	Charged the very-large rate per item, instead of the dry-goods rate.
livestock	Livestock or plants. Needs a bag count; the bag size comes from the product.
livestock_standard	Livestock, and the class itself says standard packing bag.

Meaning	What it does
livestock_large	Livestock, and the class itself says large packing bag.
collection_only	Cannot be couriered. Withholds the courier option for the whole order.
quote_required	Priced as normal, but the customer is told a quotation may be needed.
livestock_unavailable	Livestock that cannot currently be shipped at all. Withholds the courier option.

The shipped default covers both the terms the catalogue uses today and the seven target terms for when it is rebuilt:

```
# Live terms, as the catalogue stands today.
livestock = livestock
large-heavy-items = very_large
tanks-or-bubble-wrap-items = very_large
normal-products = dry_goods
local-pickup = collection_only

# The seven target terms, for when the catalogue is rebuilt.
normal-dry-goods = dry_goods
very-large-dry-goods = very_large
livestock-plants-standard-packing-bag = livestock_standard
livestock-large-packing-bag = livestock_large
collection-only = collection_only
shipping-quote-required = quote_required
livestock-shipping-unavailable = livestock_unavailable
```

Products with no shipping class (default `dry_goods`) decides what happens to a product that has no shipping class set at all. On most catalogues that is the majority of products, so the default matters: leaving it on `dry_goods` means installing the plugin does not silently reprice thousands of items on day one.

Verify your slugs

A slug that does not match any real shipping class is not an error. Those products just quietly fall through to the unclassified default. Check the actual slugs under **Products** → **Shipping classes** and correct the list. It is a two-minute job and it is the single most likely cause of a product being priced wrongly.

3.6 Checkout options and notices

Field	Default	Meaning
Offer collection from the Riverbend store	ticked	Adds a collection option with no transport charge. Packaging is still charged.
Courier option label	Courier delivery	What the courier option is called at checkout.

Field	Default	Meaning
Collection option label	Collection from our Riverbend store	What the collection option is called.
Dispatch-day notice	see below	Shown on every cart with items in it.
Notice when a quotation is needed	see below	Shown when the cart holds a <code>quote_required</code> item.
Notice when an item cannot be couriered	see below	Shown when the cart holds a <code>collection_only</code> or <code>livestock_unavailable</code> item.
Notice for outlying addresses	see below	Shown when the delivery postcode is on the outlying list.

The shipped default notices read:

Dispatch days

Livestock and plants are dispatched Monday to Wednesday only. Dry goods are normally dispatched Monday to Friday. Couriers do not normally operate over weekends. Mixed orders may be split into separate livestock and dry-goods consignments.

Quotation needed

One or more items in your order need an additional delivery quotation. We will contact you to confirm the delivery cost before dispatch.

Cannot be couriered

One or more items in your order cannot be couriered. They are available for collection from our Riverbend store.

Outlying address

Your address is in an outlying delivery area. You can place the order now; we will confirm any additional delivery fee with you before dispatch, and you are free to cancel if it does not suit you.

Clearing a notice to blank switches it off. An empty notice box is never shown.

Setting up products

4.1 The three things a product needs

1. A **WooCommerce shipping class**, which the translation list (Section 3.5) turns into a meaning. This is WooCommerce's own field, not something the plugin adds.
2. **Packing detail**, for livestock only: how many share a bag, and what size bag.
3. A **free-shipping exclusion flag**, for the products that should never earn free delivery regardless of order value.

4.2 The fields the plugin adds

On a product, these appear under **Product data** → **Shipping**.

Field	Applies to	What it means
Units per packing bag	Livestock only	How many of this product ship in one bag: three shrimp to a bag, one koi to a bag. Leave blank for anything that is not livestock.
Packing bag size	Livestock only	Standard or Large. A large bag takes twice the cooler-box space of a standard one.
Excluded from free shipping	Any product	This product never counts towards free shipping and disqualifies the whole order, whatever it is worth.

4.2.1 What happens when they are not set

Missing value	What the plugin does
Units per bag, on a livestock product	Assumes one per bag and writes a warning to the log. This over-quotes packaging rather than under-quoting it. The shop is not left absorbing the difference, but the customer may be charged for more boxes than the order needs.
Bag size, on a livestock product	Uses the size implied by the shipping class if there is one; otherwise assumes standard and writes a warning.
Shipping class, on any product	Uses whatever Products with no shipping class is set to, which is <code>dry_goods</code> by default.

Variations

Variations have their own *Units per packing bag* and *Packing bag size* fields, under the variation's dimensions. Leaving a variation's field blank makes it fall back to the parent product's value, so a value set once on the parent applies to every variation that does not override it.

4.3 Bulk editing

Setting units-per-bag one product at a time on a catalogue of several thousand items is not a task anybody finishes. All three fields are therefore available in WooCommerce's bulk edit:

1. Go to **Products**, filter to the group you want (by category, or by shipping class).
2. Tick the products, choose **Bulk actions** → **Edit**, and select **Apply**.
3. The three fields appear at the bottom of the bulk-edit panel.

Every field defaults to – No change –, so selecting four hundred products to set one value cannot silently blank the other two.

4.4 A practical order of work

1. Confirm the shipping-class slugs and fix the translation list.
2. Bulk-set *Excluded from free shipping* on the categories that need it (tanks, hardscape, wood, rocks, and any named exclusions).
3. Filter to the livestock shipping class and bulk-set *Packing bag size* to Standard.
4. Filter to the large-bagged species and bulk-set them to Large.
5. Bulk-set *Units per packing bag* group by group: shrimp and small tetras at several per bag, cichlids and koi at one.
6. Check the log (Chapter 7) for any livestock product the plugin had to guess at.

How the price is calculated

The calculation runs in six steps, in a fixed order. This chapter follows that order. Everything here is deterministic: the same cart with the same settings always produces exactly the same answer.

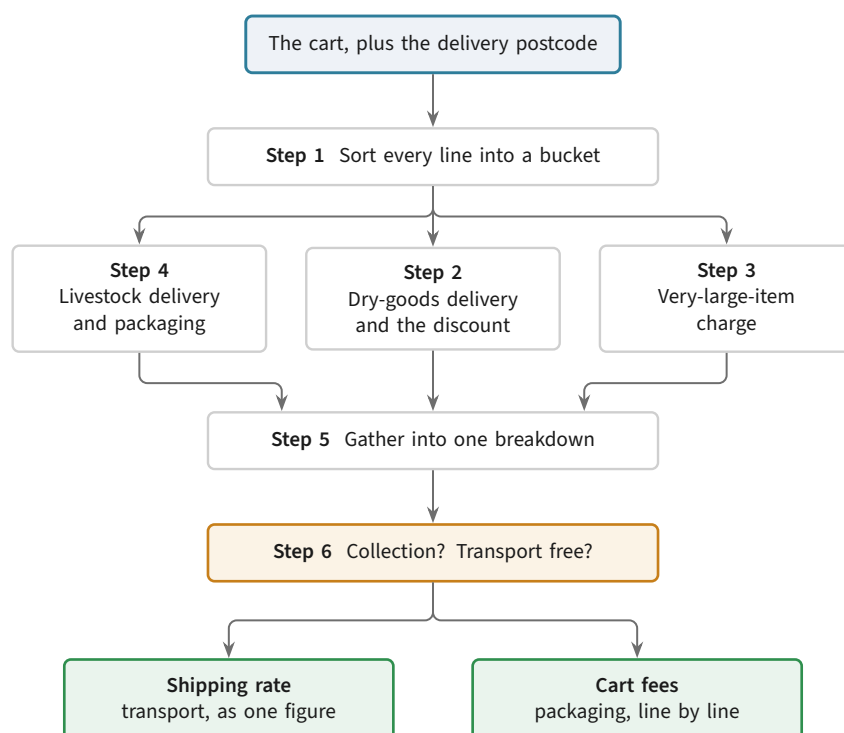


Figure 5.1. The whole calculation. Steps 2, 3 and 4 are independent of one another. None can affect the others' figures.

5.1 Step 0: which area is this going to?

Before anything is priced, the delivery postcode is classified.

Area	When
Outlying	The postcode is on the outlying list.
Local	The postcode is on the local list.
National	Everything else: a postcode on neither list, a blank postcode (as on the cart page before an address is entered), or an address outside the country.

Only livestock delivery has separate local and national rates. Outlying is priced identically to National; the difference is that the order is flagged for a human to confirm any additional fee.

Note

Falling back to National for an unknown postcode is the safe direction to be wrong in. Quoting R280 where R120 was due is a refund; quoting R120 where R280 was due is a loss the shop absorbs on every order until somebody notices.

5.2 Step 1: sorting the cart

Every line in the cart is put into exactly one bucket, based on its shipping class:

Bucket	Contents
Dry goods	Everything mapped to <code>dry_goods</code> . The bucket keeps a running subtotal and a line count.
Very large	Everything mapped to <code>very_large</code> . The bucket keeps a total <i>quantity</i> , not a line count. Two of the same aquarium is two items.
Livestock	Everything mapped to a livestock meaning. Lines pointing at the same product are merged and their quantities summed <i>before</i> any bag arithmetic.
Flagged	<code>collection_only</code> , <code>quote_required</code> and <code>livestock_unavailable</code> . These never produce a charge; they carry a restriction, not a price.

At the same time, two figures are collected for the free-shipping decision: the **gross merchandise value** (the sum of every line's subtotal, before coupon discounts and excluding tax) and whether **any product carries the exclusion flag**.

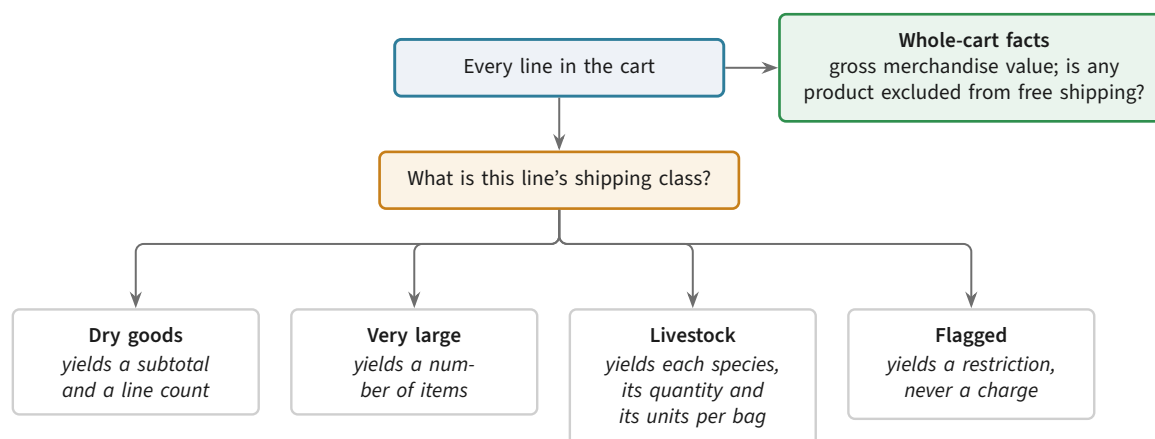


Figure 5.2. Step 1. Each line goes into exactly one bucket. The two whole-cart facts are collected alongside, and feed the free-shipping test rather than any bucket.

5.3 Step 1a: does this order earn free shipping?

The order qualifies only if **all three** are true:

1. The gross merchandise value meets the threshold. (Whether “meets” means $\geq$ or $>$ is the *exactly on the threshold* setting.)

2. No shipping class in the cart is on the blocking list.
3. No product in the cart is individually flagged as excluded from free shipping, either by its tick-box or by exceeding the weight limit.

If it does not qualify, the plugin also works out the *shortfall*, how much more merchandise would meet the threshold, so the cart can tell the customer.

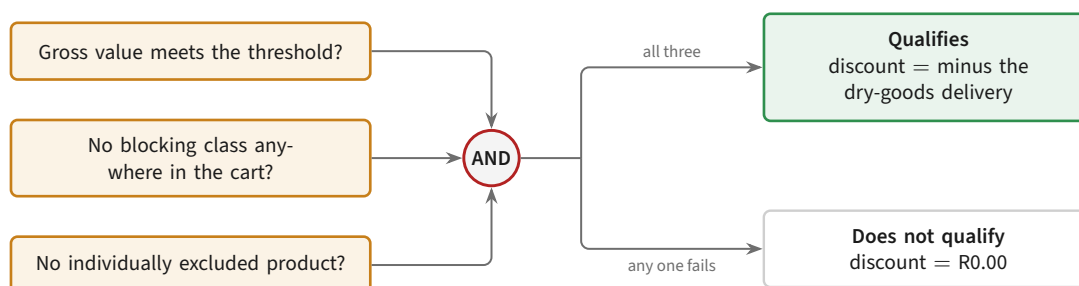


Figure 5.3. The free-shipping test. Three independent conditions, all of which must hold.

Important

Because livestock and very-large items are both on the blocking list by default, a R5 000 order containing a single neon tetra does not earn free shipping. That is deliberate: one bag of fish still costs a cooler box and a courier run. It is also stricter than a threshold that gates on order value alone, so it is worth confirming before go-live that this is the behaviour the shop wants customers to meet.

5.4 Step 2: dry-goods delivery

If the dry-goods bucket has anything in it, two lines are produced:

- **Dry-goods delivery:** the flat rate. One charge for the whole bucket, not per item.
- **Free-shipping discount:** exactly minus the delivery charge if the order qualifies, otherwise R0.00.

Both lines appear even when the discount is zero, because Free-shipping discount: R0.00 next to Dry-goods delivery: R160.00 tells the customer something useful. The discount exists and this order did not earn it.

The discount is always computed from the same delivery charge it cancels, never from a second setting, so the two can never drift apart.

If the cart has *no* dry goods in it (a livestock-only order, say), neither line appears at all. There is no dry-goods delivery to charge.

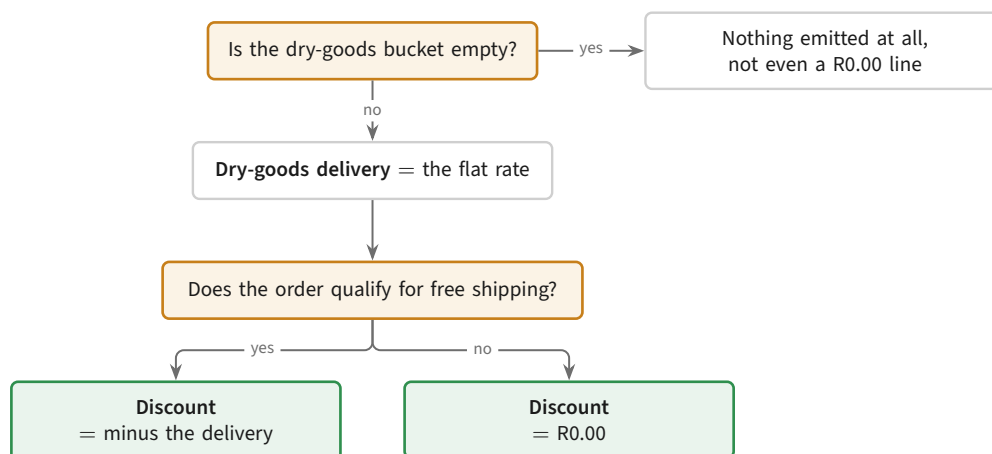


Figure 5.4. Step 2. Both lines are emitted together whenever there are dry goods, even when the discount comes to nothing.

5.5 Step 3: very-large items

If the very-large bucket has anything in it:

$$\text{Very-large-item charge} = \text{very-large rate} \times \text{number of very-large items}$$

The item count is recorded alongside the money, so a R640 line visibly comes from two items rather than being an unexplained number.

Very-large products never enter the dry-goods bucket, so they never attract the dry-goods rate as well. The specification says the very-large charge is used *instead of* the normal rate, and this is where that is enforced.

5.6 Step 4: livestock delivery and packaging

Two halves that run independently. The delivery rate depends only on where the order is going; the packaging depends only on what is in it.

5.6.1 Delivery

One **Livestock delivery** charge for the whole consignment, at the local or the national rate.

5.6.2 Bags

For each livestock product *separately*:

$$\text{bags} = \left\lceil \frac{\text{quantity}}{\text{units per bag}} \right\rceil$$

The $\lceil \rceil$ brackets mean *round up to the next whole number*. Ten fish at three to a bag is four bags, not three and a third.

Products are counted on their own before anything is summed, because a bag never mixes species. Four shrimp and one betta are five animals but two bags.

5.6.3 Cooler boxes

Bags from every product are then pooled, because standard and large bags *may* share a cooler box:

$$\text{slots} = \text{standard bags} + (\text{large bags} \times \text{large-bag slots})$$

$$\text{cooler boxes} = \left\lceil \frac{\text{slots}}{\text{cooler-box capacity}} \right\rceil$$

With the shipped defaults that is $\lceil (\text{standard} + \text{large} \times 2) / 4 \rceil$.

5.6.4 Charges

- **Cooler-box charge** = charge per box × number of boxes.
- **Heat-pack charge** = charge per pack × number of boxes, but *only* while heat-pack season is on. One heat pack per cooler box.

The box count and the heat-pack count are both recorded and displayed.

An order with no cooler boxes produces no packaging lines at all, not a pair of R0.00 lines. There is no packaging to account for, so there is nothing to say.

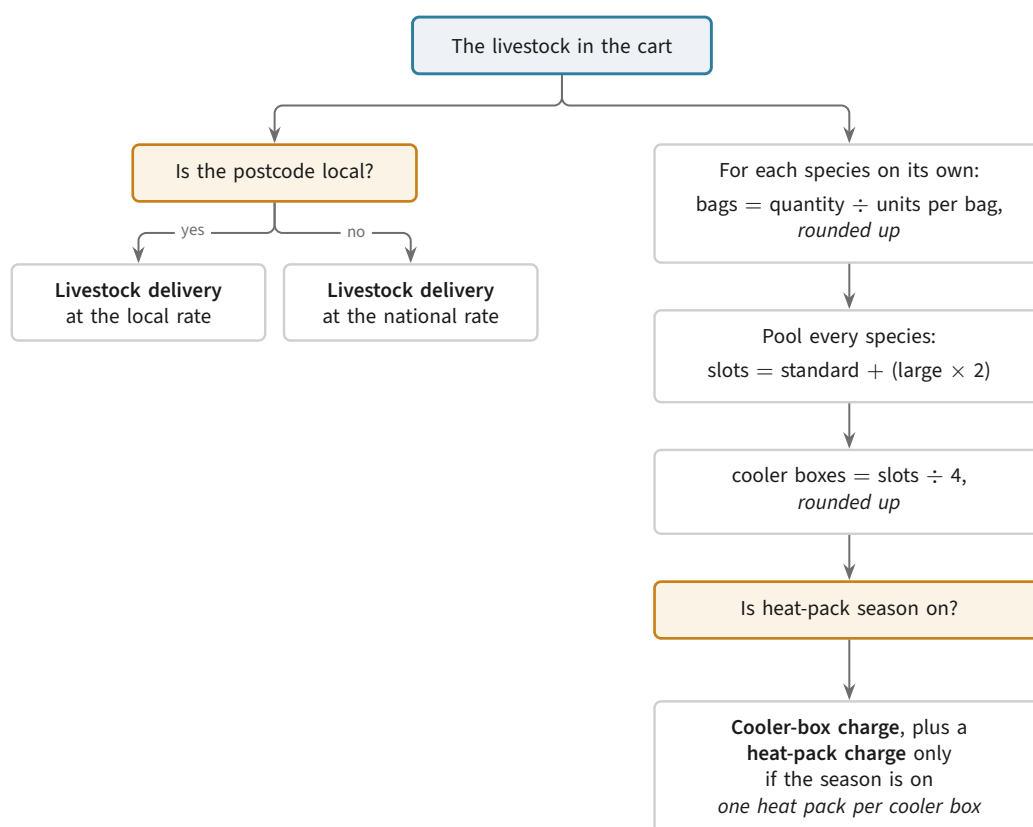


Figure 5.5. Step 4. The two halves are independent: where the order is going decides the delivery rate, what is in it decides the packaging.

Note

Nothing in this step consults the free-shipping decision. Livestock orders never earn free shipping, and that stays true by construction rather than by a rule that could be edited away.

5.7 Step 5: gathering the breakdown

The three sets of charges are merged into one breakdown, in a fixed display order:

1. Livestock delivery
2. Dry-goods delivery
3. Cooler-box charge
4. Heat-pack charge
5. Very-large-item charge
6. Free-shipping discount
7. Manual surcharge or adjustment

Each line may appear at most once. An attempt to add the same line twice is a hard error rather than a silent double charge.

Any flagged items are attached at this point, and so is a manual adjustment if one has been set (see Section 10.5).

5.8 Step 6: the whole-order rules

Two rules that act on the order as a whole, in this order:

1. Collection removes transport. If the customer is collecting, every transport line is zeroed: livestock delivery, dry-goods delivery, the very-large charge, *and* the free-shipping discount. Packaging is untouched and stays payable.

Note

The free-shipping discount is treated as transport for exactly this reason. If collection zeroed the R160 delivery but left a —R160 discount standing, a collected order would quote *minus* R160, the shop paying the customer to come and fetch their own order.

2. If transport ended up free, packaging may follow. If the order had transport charges and they came to nothing, and *Packaging stays payable when transport is free* is unticked, the packaging is zeroed too. With the default setting (ticked) this rule never fires.

A manual adjustment survives both rules, because it is neither transport nor packaging.

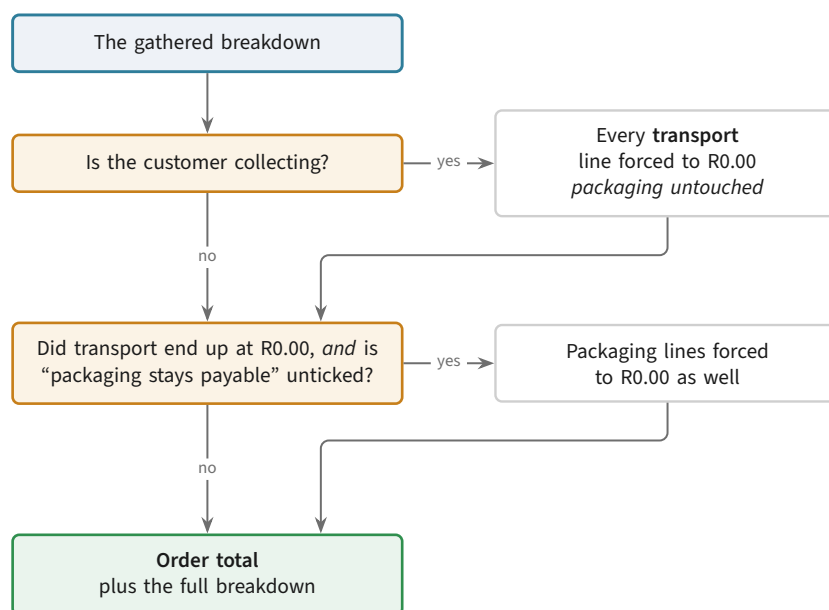


Figure 5.6. Step 6. With the shipped default (packaging stays payable), the second rule never fires.

5.9 Where the money ends up at checkout

The result is split across two WooCommerce mechanisms, and the split matters:

Shown as	Carries
The shipping rate	Livestock delivery, dry-goods delivery, the very-large charge and the free-shipping discount, added together into one figure.
Cart fees	The cooler-box charge, the heat-pack charge and any manual adjustment, each as its own line.

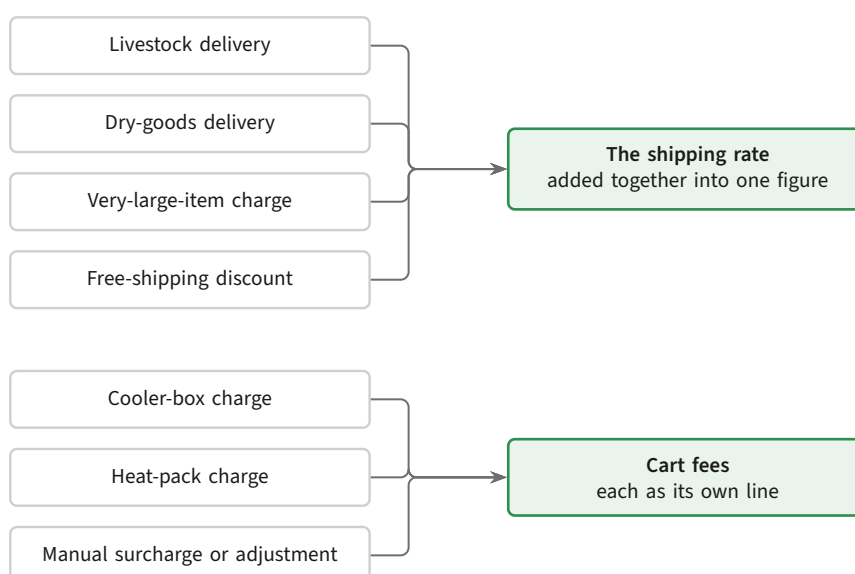


Figure 5.7. Where each component lands at checkout. The split follows the cost line's own definition, not a hand-maintained list.

This is not a stylistic choice. A shipping rate disappears when the customer picks a different shipping method; a fee does not. Because packaging has to survive a switch to collection, packaging has to be a fee.

5.10 Worked example: a mixed order

Cart, delivered nationally, with heat-pack season on:

Item	Class	Qty	Line total
Aquarium light	Dry goods	1	R800.00
Glass aquarium	Very large	2	R2 400.00
Cherry shrimp (3 per bag)	Livestock	13	R390.00
Gross merchandise value			R3 590.00

Free shipping: the gross value is over R2 500, but the cart contains both very-large and livestock items, which are both on the blocking list. The order does **not** qualify. Discount R0.00.

Bags: $13 \div 3 = 4.33$, rounded up to 5 standard bags. **Boxes:** $5 \div 4 = 1.25$, rounded up to 2 cooler boxes. **Heat packs:** 2, because the season is on.

Component	Amount
Livestock delivery	R280.00
Dry-goods delivery	R160.00
Cooler-box charge ($2 \times R60.00$)	R120.00
Heat-pack charge ($2 \times R25.00$)	R50.00
Very-large-item charge ($2 \times R320.00$)	R640.00
Free-shipping discount	R0.00
Total shipping and packaging	R1 250.00

Of that, R1 080.00 appears as the shipping rate and R170.00 as two cart fees.

5.11 Worked example: a small local order

Cart, delivered to a local postcode, heat-pack season off:

Item	Class	Qty	Line total
Super Glue	Dry goods	1	R35.00
Shallow desktop aquarium	Very large	1	R600.00
Assorted angelfish (1 per bag)	Livestock	2	R140.00
Merchandise subtotal			R775.00

Bags: 2 standard. **Boxes:** $2 \div 4 = 0.5$, rounded up to 1. **Heat packs:** 0, season off.

Component	Amount
Livestock delivery (local)	R120.00
Dry-goods delivery	R160.00
Very-large-item charge	R320.00
Free-shipping discount	R0.00
<i>Courier delivery (shipping rate)</i>	<i>R600.00</i>
Cooler-box charge (fee)	R60.00
Merchandise subtotal	R775.00
Order total	R1 435.00

If the same customer switched to collection, the R600.00 would become R0.00 and the R60.00 cooler-box fee would remain, for an order total of R835.00.

What customers and staff see

6.1 On the cart and checkout

6.1.1 The delivery options

Up to two options are offered: the courier option and, if enabled, the collection option. Both are labelled from the settings.

If the cart contains anything that cannot be couriered, a collection-only item or livestock the shop cannot currently ship, then the **courier option is withheld entirely** and collection is offered even if it is normally switched off, so the customer still has a way to complete the order. Quoting a courier price for an order that cannot go on a courier, and discovering it after payment, is the failure this prevents.

6.1.2 The breakdown under the rate

When the transport figure is made up of more than one component, those components are listed under the rate at checkout. A single component is not listed, because repeating one number under two labels explains nothing.

Zero lines are hidden, with one exception: the free-shipping discount is always shown when it is present, because Free-shipping discount: R0.00 is genuinely informative.

6.1.3 The packaging fees

Cooler boxes and heat packs appear as their own cart fee lines. When there is more than one, the count is put in the label so the customer can check the arithmetic:

Cooler-box charge (2 x R60.00) R120.00

6.1.4 The notices

Up to four notices are shown on the cart, in this order: the dispatch-day notice (always, on any non-empty cart), then the quotation notice, the cannot-be-couriered notice and the outlying-address notice as they apply.

Notices do not show on the block cart or checkout

The notices are added using WooCommerce's standard notice mechanism. That works on the **classic** shortcode cart and checkout pages. It does **not** work on the newer **block** cart and checkout, which render their own notices and discard informational ones added this way. Pricing is completely unaffected: this is display only. If the store uses block checkout pages and the notices are needed, supporting them means shipping a small JavaScript bundle through the WooCommerce Blocks integration API. See [Chapter 9](#).

6.2 On the order

6.2.1 The customer's view

On the order-received page and in every order email, the breakdown is inserted directly *underneath* the shipping line, where it explains the number above it.

Only components that add something are shown. Specifically:

- Zero-value lines are omitted.
- Packaging and manual adjustments are omitted, because they are already real fee line items that WooCommerce prints itself, and printing them again would say Cooler-box charge: R60.00 twice.
- If only *one* transport component remains, nothing is shown at all: one line on its own is just WooCommerce's own Shipping row under a different name. Two or more genuinely explain how the figure was reached.

The cooler-box, heat-pack and very-large-item *counts* are shown alongside.

6.2.2 The shop's view

The same breakdown appears on the order edit screen in wp-admin, under the shipping total, plus one extra row, Needs attention, listing any flagged items.

The plugin also adds **order notes** for anything a person has to act on:

- An outlying delivery address, with a reminder to confirm the additional fee before dispatch.
- A list of items needing attention before dispatch.
- A warning that the order contains items that cannot be couriered.

6.2.3 Why the numbers never change retrospectively

The breakdown is written onto the order as it is created, and it is *read back* from the order rather than recalculated. An order placed in March still shows the figures it was actually charged, not what today's settings would produce.

Routine maintenance

7.1 Changing a rate

Go to the settings screen, change the figure, save. That is the whole procedure. Every rate is in exactly one place, so a rate can never drift out of step with a copy of itself.

Existing orders are unaffected. They keep the figures they were charged.

Important

Remember the figure you type excludes VAT. If shipping is taxable on this store, the customer pays more than the number you enter.

7.2 Turning heat-pack season on and off

In **by hand** mode (the default): tick or untick *Heat-pack season is on now* and save. The change takes effect on the next cart calculation.

To automate it, set *Heat-pack season* to Between two dates each year and set the start and end to MM-DD values. The manual tickbox is then ignored.

7.3 Adjusting the local delivery area

Edit the *Local delivery postcodes* box. Use ranges wherever possible. A textarea holding several hundred four-digit numbers is a setting nobody maintains correctly.

If you add a postcode that is already on the outlying list, the save is rejected and the conflicting entries are named.

7.4 Adding a new product

1. Set the WooCommerce shipping class as normal.
2. If it is livestock: set *Units per packing bag* and *Packing bag size*.
3. If it should never earn free shipping: tick *Excluded from free shipping*.
4. Add it to a test cart and check the quoted figure before it goes live.

7.5 Reading the log

When the plugin has to guess at something, it says so, without ever interrupting a checkout. Warnings go to:

WooCommerce → **Status** → **Logs**, source `riverbend-shipping`

Typical entries and what to do about them:

Message	What to do
“...” is livestock but has no units-per-bag set. Assuming 1 per bag	Set <i>Units per packing bag</i> on that product. Until you do, its packaging is over-quoted.
“...” is livestock but has no packing-bag size set	Set <i>Packing bag size</i> , or map its shipping class to <code>livestock_standard / livestock_large</code> .
Product ... is in a cart but could not be loaded	The product was deleted while in somebody's cart. It was treated as dry goods. Usually harmless.
Shipping settings are invalid, so the shipped defaults are being used	Act on this immediately. A setting is unreadable and the shop is quoting default prices. Open the settings screen; it will name the problem.

Each distinct message is logged once per page load rather than once per cart line, so the log stays readable.

7.6 Checking a quote by hand

To reproduce a figure a customer was quoted:

1. Open the order and read the stored breakdown, which records exactly what was charged.
2. If the order looks wrong, rebuild the same cart on the front end and compare.
3. If the two differ, a setting or a product's data has changed since the order was placed. The order is right; the current settings are what changed.

7.7 Upgrading the plugin

1. Take a backup, or work on staging first.
2. Go to **Plugins**, deactivate **Riverbend Shipping** and delete it.
3. Upload and activate the new zip.

Deleting the plugin through wp-admin does not remove its settings or any order data, so settings survive an upgrade. Adding a setting in a later version does not break an existing install. An unsaved new setting simply uses its default.

Troubleshooting

Symptom	Likely cause	Fix
No Riverbend option appears at checkout at all.	The method has not been added to the shipping zone the address falls into.	WooCommerce → Settings → Shipping → Zones , add Riverbend Shipping to the zone.
An admin notice says WooCommerce is missing.	WooCommerce is deactivated or not installed.	Activate WooCommerce. The plugin does nothing without it.
A product is charged the dry-goods rate when it should be very large.	Its shipping-class slug is not in the translation list, or is spelled differently.	Compare Products → Shipping classes against the translation list and correct the slug.
Packaging is charged on an order with no livestock in it.	A non-livestock product has been mapped to a livestock meaning.	Check that product's shipping class and the translation list.
Too many cooler boxes are quoted.	Livestock products have no units-per-bag set, so each animal is being given its own bag.	Set <i>Units per packing bag</i> . The log names every product affected.
Free shipping is never offered.	Something in the cart is on the blocking list, or a product is individually excluded, or exceeds the weight limit.	Check the blocking-class list, the product's exclusion tickbox, and its weight against the limit.
Free shipping is offered when it should not be.	The relevant class is not on the blocking list, or the product is not flagged.	Add the class to <i>Shipping types that block free shipping</i> , or tick the product.
The courier option has vanished.	The cart contains a collection-only or livestock-unavailable item.	This is intended. The customer is shown the reason via the notice, unless the site uses block checkout.
Notices never appear.	The site uses the block cart/checkout.	See Section 6.1.4. Pricing is unaffected.
Customers are paying more than the rate entered.	The rates are ex-VAT and shipping is taxable.	Either lower the rate to the ex-VAT equivalent, or change the shipping tax status.
No breakdown appears on the order.	There is only one transport component, so there is nothing to explain.	Intended. The WooCommerce Shipping row already shows that figure.

Symptom	Likely cause	Fix
The whole shop is suddenly quoting default prices.	A saved setting is invalid, so the plugin fell back to defaults.	Open the settings screen, which names the violation, and check the log.
Two collection options are shown.	Both this plugin's collection option and WooCommerce Local Pickup are enabled.	Turn off one of them. See Section 9.3 .

Known limitations and open decisions

Everything in this chapter works as described. These are places where a decision has been made on the shop's behalf and is worth confirming, or where a known gap exists.

9.1 Decisions made on your behalf

Question	What the plugin does today
Does exactly R2 500 qualify for free shipping?	Yes, inclusive. Reversible with one tickbox.
Should packaging appear as money on the invoice?	Yes: two lines, Cooler-box charge and Heat-pack charge. Without them, packaging money appears in the total with nothing accounting for it.
Is free shipping automatic or opt-in?	Automatic. It is applied the moment the order qualifies, rather than offered as an option the customer has to notice and select.
How are livestock grouped into bags?	By product, not by species. Two different shrimp products each get their own bag, even though a packer would combine them. Closing this needs a species field, which is a catalogue change rather than a plugin one.
Are the free-shipping exclusions actually enforced?	Yes. Both the blocking classes and the per-product tickbox are hard gates, not advisory. An order failing either one is never offered free delivery.
The over-4 kg exclusion.	On by default, and adjustable. It only acts on products that carry a weight, so on a catalogue with patchy weights it applies unevenly. Set the limit to 0 to switch the rule off entirely.

9.2 Still outstanding

The local-delivery postcode list. What ships is a placeholder boundary, not a verified one. It has to be checked against the areas the shop actually drives to before go-live.

Units per bag, per species. No default is assumed. Every livestock product needs this set before its packaging can be quoted accurately.

A local tier for dry goods and very-large items. Only livestock has a local rate. If a cheaper local rate is wanted for the other two, it is a settings change plus one line of code, with no change to the pricing logic.

The shipping-class slugs. The seeded translation list has to be checked against the slugs the store really uses. A wrong slug does not error. Those products silently fall through to the unclassified

default.

Notices on the block cart and checkout. Section [6.1.4](#). This matters the day anybody rebuilds those pages with blocks.

9.3 Collection appears twice if Local Pickup is also enabled

The plugin offers its own collection option, and it also recognises WooCommerce's built-in Local Pickup as collection, and both correctly remove transport while keeping packaging payable.

If both are configured, customers see two collection options. Pick one: either untick *Offer collection from the Riverbend store* in the plugin settings, or remove Local Pickup from the shipping zone.

Technical reference

This chapter is for whoever maintains the code. It is not needed to run the shop.

10.1 What is in the distributed package

Path	Contents
<code>riverbend-shipping.php</code>	Plugin header and bootstrap. Registers the autoloader and boots the plugin.
<code>includes/</code>	The pricing engine. No WordPress code.
<code>wordpress/</code>	The WooCommerce adapter.

Nothing else ships. There is no `vendor` directory: the plugin registers its own PSR-4 autoloader, so Composer is a development tool only.

10.2 Building a release

From the plugin repository:

```
./build-release.ps1
```

This reads the `Version:` header from `riverbend-shipping.php`, stages the three runtime paths into a folder named after the plugin slug, and writes `build/riverbend-shipping-<version>.zip`. Bumping the version header is the only step needed before cutting a new build.

10.3 Development tooling

```
composer install
composer check
```

`composer check` runs, in order: `phpcs` (PSR-12), `phpstan` at level 9 over both trees, and `phpunit`. They can be run individually as `composer cs`, `composer stan` and `composer test`.

PHPStan uses two configurations. One covers the engine; the other covers the adapter and loads the WordPress and WooCommerce stubs. The split is deliberate: the stubs are deliberately *not* available to the engine, which is what makes “no WordPress code in the engine” something a tool can verify rather than a habit.

10.4 Design rules worth preserving

- **Money is integer cents.** The engine never divides money, so there is no rounding policy to get wrong.

- **Exactly one ceiling function.** Both the bag count and the cooler-box count go through it, and it refuses floats outright.
- **Rates are VAT-exclusive.** VAT is WooCommerce's job.
- **Purity.** The same cart plus the same settings always produces an identical quote. There is no clock in the engine: heat-pack season is resolved to a yes or no at the boundary, before the engine is built.
- **The transport/packaging split follows the cost line's own definition,** not a hand-maintained list. A new cost line lands on whichever side its definition puts it.

10.5 The manual-adjustment filter

The breakdown supports a Manual surcharge or adjustment line, but no pricing rule produces one: it is a human decision about a specific order. Something else can set it through a filter, in integer cents:

```
add_filter( 'riverbend_shipping_manual_adjustment_cents', function ( $cents ) {  
    return 6000; // R60.00 surcharge  
} );
```

A manual adjustment is added as a cart fee and survives both whole-order rules in Section 5.8.

10.6 Verification status

At version 0.1.0 the plugin has been exercised in two ways:

- An **automated suite** covering the engine, the adapter's cart mapping, the settings parsing and a set of end-to-end pipeline scenarios, including the worked examples reproduced in Chapter 5.
- A **27-scenario manual test round** against a staging store with a real product catalogue, covering every shipping class, both fulfilment choices, free-shipping boundaries, packaging arithmetic, notices, order storage and the settings screen. All 27 passed; the only finding was the block-checkout notice limitation in Section 6.1.4.

Settings quick reference

Every option is stored with the prefix `riverbend_shipping_`.

Setting	Option key	Default
<i>Shipping rates</i>		
Local livestock delivery	<code>rate_livestock_local</code>	120.00
Livestock delivery, national	<code>rate_livestock_national</code>	280.00
Normal dry-goods delivery	<code>rate_dry_goods</code>	160.00
Very large product delivery	<code>rate_very_large</code>	320.00
<i>Free shipping</i>		
Threshold	<code>free_shipping_threshold</code>	2500.00
Threshold is inclusive	<code>free_shipping_threshold_inclusive</code>	yes
Blocking shipping types	<code>free_shipping_blocking_classes</code>	all five
Weight limit (kg)	<code>free_shipping_weight_limit_kg</code>	4
<i>Packaging</i>		
Bags per cooler box	<code>cooler_box_capacity</code>	4
Slots used by a large bag	<code>large_bag_slots</code>	2
Charge per cooler box	<code>cooler_box_charge</code>	60.00
Charge per heat pack	<code>heat_pack_charge</code>	25.00
Season mode	<code>heat_pack_season_mode</code>	manual
Season on now	<code>heat_pack_season_active</code>	no
Season start	<code>heat_pack_season_start</code>	05-01
Season end	<code>heat_pack_season_end</code>	08-31
Packaging on free transport	<code>packaging_payable_on_free_courier</code>	yes
<i>Areas</i>		
Local postcodes	<code>local_postcodes</code>	Riverbend ranges
Outlying postcodes	<code>outlying_postcodes</code>	empty
<i>Catalogue</i>		
Shipping class translations	<code>shipping_class_map</code>	12 rules
Unclassified meaning	<code>unclassified_meaning</code>	dry_goods
<i>Checkout</i>		
Offer collection	<code>offer_collection</code>	yes
Courier label	<code>courier_label</code>	Courier delivery
Collection label	<code>collection_label</code>	Collection from our...
Dispatch-day notice	<code>notice_dispatch_days</code>	see text

Setting	Option key	Default
Quotation notice	notice_quote_required	see text
Undeliverable notice	notice_undeliverable	see text
Outlying-area notice	notice_outlying_area	see text

Data the plugin stores

B.1 Product meta

Key	Field	Values
<code>_rb_units_per_bag</code>	Units per packing bag	whole number ≥ 1 , or blank
<code>_rb_bag_type</code>	Packing bag size	standard, large, blank
<code>_rb_free_shipping_excluded</code>	Excluded from free shipping	yes / no

The shipping class itself is WooCommerce's own `product_shipping_class` taxonomy, reused rather than duplicated.

B.2 Order meta

Key	Holds
<code>_rb_livestock_delivery</code>	Livestock delivery charged, as a decimal string
<code>_rb_dry_goods_delivery</code>	Dry-goods delivery charged
<code>_rb_cooler_box_charge</code>	Cooler-box money charged
<code>_rb_heat_pack_charge</code>	Heat-pack money charged
<code>_rb_very_large_charge</code>	Very-large-item money charged
<code>_rb_free_shipping_discount</code>	Discount applied (negative, or zero)
<code>_rb_manual_adjustment</code>	Manual surcharge or adjustment
<code>_rb_cooler_boxes_qty</code>	Number of cooler boxes
<code>_rb_heat_packs_qty</code>	Number of heat packs
<code>_rb_very_large_items_qty</code>	Number of very-large items
<code>_rb_delivery_area</code>	local, national OR outlying
<code>_rb_fulfilment</code>	courier OR collection
<code>_rb_requires_quote</code>	yes / no
<code>_rb_needs_area_confirmation</code>	yes / no
<code>_rb_flagged_items</code>	Newline-separated list of items needing attention

One row per component, so a question like “how much did we take in cooler-box charges last month” can be asked of the database directly.

Glossary

Term	Meaning
Bag	One packing bag of livestock. Never mixes species.
Blocking class	A shipping type whose presence anywhere in the cart disqualifies the whole order from free shipping.
Bucket	One of the four groups the cart is sorted into: dry goods, very large, livestock, flagged.
Cooler box	An insulated box holding several bags. Charged per box.
Cost line	One named component of the shipping figure.
Flagged item	An item carrying a restriction rather than a price: collection-only, quote-required, or livestock-unavailable.
Fulfilment	Whether the order is couriered or collected.
Gross merchandise value	The value of everything in the cart before coupon discounts, excluding shipping, packaging and tax. What the free-shipping threshold is measured against.
Large bag	A packing bag using twice the cooler-box space of a standard one.
Outlying	A postcode priced at the national rate but flagged for a human to confirm an additional fee.
Shipping class	WooCommerce's own product taxonomy, translated into the plugin's vocabulary by a setting.
Slot	A standard-bag space in a cooler box. A large bag uses two.
Transport	The money charged for moving an order, as opposed to for packing it. Removed by collection; packaging is not.